OpenADMET CYP450 Prediction

Team: Hamza Habib, Yuanqing Wang

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Model Details

We submit a per-isoform ensemble, 4-5 members each, whose members are chosen to differ in representation and in how they use the available data. We utilize the following learners:

1. Isoform-conditioned multi-task Chemprop network built on a CheMeleon-pretrained encoder that predicts all four isoforms through one shared head conditioned with an ESM-2-derived isoform code, trained with credible-interval row weights, auxiliary heads for single-dose log2 fold-change, external pIC50, and binary active/inactive heads
2. A gradient-boosted model on fingerprint and molecular descriptor bank, with a multi-task variant for CYP2C9
3. A two-stage low-to-high-fidelity model in which out-of-fold predictions of the single-dose log2 fold-change from several encoders (Chemprop, Uni-Mol2, Boltz-2 embeddings) are fed with RDKit descriptors to a TabPFN readout
4. A SuperCowPowers-style family of Chemprop models
5. Isoform-specific additions such as an CYP1A2 specific multitask learner trained without CYP2D6 model and a CYP2D6 low-potency specialist.

Members are averaged with equal weights and the predictions are then affinely recalibrated per isoform.

Data Details

Primary training data comes from the OpenADMET CYP inhibition blind challenge with 4,905 compounds in a sparse compound-by-isoform matrix of direct-inhibition pIC_{50} labels for each of the isoforms along with lower and upper bounds from the dose-response fit.
Single-concentration primary screening data for 17,504 compound-enzyme readings across 4,376 compounds were provided in the form of $\log_2$ fold-change (log2fc) of the readout against the control.

The structures of the 750 blind compounds were used to generate end-point predictions, determine similarity statistics and select label-free auxiliary structures. Additional $E_{\max}$ and time-dependent inhibition (TDI) information was released for the compounds.
We used the following public datasets: NCGC qHTS deposits, AID-1851 PubChem Assay, Tox21 P450-Glo, ChEMBL-37, ADME@NCATS, Novartis surrogate ADME predictions.